SolvePath

Agent architecture · one tutor turn

A fixed Google ADK pipeline between two short Postgres transactions. Deterministic agents decide; LLM agents only classify and phrase. No model chooses a tool, route, verdict, hint level or state transition.

Deterministic · LLM · schema-bound · MCP client · A2A client

TX-A · open — POST /events → auth on `sub` → rate limit (60/h, 15/5 min) → idempotency (24 h) → **session lease** (one in-flight event) → attempt stored. Connection released before any MCP, A2A or LLM call.

run_turn → SequentialAgent

PHASE 1 Understand the attempt

deterministic + MCP · no model calls

AG-01 · DETERMINISTIC — Problem Context
Load item version, scrub and normalise input, run the input screen.
`question_id · version · answer_text ≤ 64 ch`
↯ Unreadable → canned rephrase

AG-02 · MCP CLIENT — Math Verifier
Verify the answer and steps against the authored solution.
`MCP verify_expression → correct · partially_correct · incorrect · cannot_verify`
⏱ 400 ms
↯ Timeout → one in-process SymPy fallback

AG-03 · DETERMINISTIC — Misconception Matcher
Match the response against authored misconception signatures.
`exact-key matcher → candidate codes`
↯ No match → defer to AG-06

AG-04 · MCP CLIENT — Retrieval Gate
Fetch similar exemplars only when policy allows.
`MCP search_exemplars · pgvector`
⏱ 250 ms
↯ Timeout → continue without exemplars

AG-05 · DETERMINISTIC — Student State Estimator
Update attempts, hint level history and skill evidence.
`learner model · tutoring_session.turn_number (cap 12)`

allow-listed ADK state keys only

PHASE 2 Decide, then phrase

policy decides · LLM phrases · guard checks

AG-06 · LLM · SCHEMA-BOUND — Misconception Classifier
Labels a likely misconception when the matcher is unsure. Never changes correctness.
`≤ 1 call · one schema repair`
⏱ 1,000 ms
↯ Invalid / timeout → no label

AG-07 · DETERMINISTIC — Hint Policy Engine
Selects hint level (L0–L5), probe and prompt type from versioned policy data.
`policy hp-1.0 · transitions.apply`
⏱ 40 ms

LOOPAGENT · CANDIDATE → CHECK → GUARD — MAX 2 ITERATIONS

AG-08 · LLM · SCHEMA-BOUND — Socratic Dialogue
Phrases only what AG-07 selected. One question per turn. Never sees the accepted answer.
`≤ 1,500 prompt / ≤ 250 completion tokens`
⏱ 1,400 ms
↯ Fail → authored, maths-free fallback

AG-09 · DETERMINISTIC — Leak Check
Deterministic answer-leak rules. A hit always beats a guard allow.
`answer_leakage = 0 (merge gate)`
↯ Hit → revise or fallback

AG-10 · A2A CLIENT — Safety Guard
Small-LLM safety verdict from the separate safety service.
`A2A → allow · block · revise · escalate`
⏱ 1,200 ms
↯ Cap 1.3 s · fail closed

↺ revise → back to AG-08 (second and final candidate)

final override · no draft stored or streamed before this point

TX-B · commit — Persist tutor_turn, state transition, learner update and an insert-only `audit_event` through the unit-of-work writer (200 ms reserve). Release lease → respond with `tutor.controls`. Langfuse flush ≤ 300 ms, never fails the turn.

SATELLITE SERVICES

`solvepath-mcp`
MCP server
Math verifier and exemplar search. SymPy, mpmath, pgvector. Bundle < 30 MB, no ADK.

Auth	HMAC bearer, ≤ 60 s, own secret
Callers	AG-02, AG-04
On failure	Timeout → one in-process fallback. 4xx/5xx never retried

`solvepath-safety`
A2A safety guard
Verdict-only service on MODEL_SAFETY. No read access to domain data.

Auth	HMAC bearer, ≤ 60 s, own secret
Callers	AG-10
On failure	Fail closed → authored fallback

`LLM adapter (L7)`
Gemini default · OpenRouter alternate
Dialogue, classifier, safety model, teacher insight, embeddings.

Auth	Provider API keys (secret store)
Callers	AG-06, AG-08
On failure	Timeout, 429, 5xx, bad schema → deterministic fallback

`Langfuse Cloud`
Tracing & prompts
One trace per turn, one span per agent step. MCP and A2A are child spans via traceparent.

Auth	Public / secret keys
Callers	Orchestrator
On failure	trace_status = missing

STAGE BUDGETS · 7.5 S HARD STOP

ver | ret | class | dialogue | guard

Sum of stage caps ≈ 4.5 s — CI gate: turn p95 < 8 s

No open agent loop
No ReAct, no model-chosen tools or routes. The only loop is AG-08 → AG-10, capped at 2.
§5.2 #1

No free text to models
Student free text never reaches a prompt, span, cache key or value in Release 1.0.
§5.2 #2

LLM agents hold nothing
No tool client, repository or accepted answer. Only allow-listed ADK state keys.
§5.2 #5

No DB across calls
Pipeline runs between TX-A and TX-B. No connection held across MCP, A2A or LLM calls.
§5.2 #17

Source: uploads/TECHNICAL.md §3.2–3.3, §4.1, §4.3, §5.1–5.2 · agent ids from the layered architecture diagram (DESIGN.md §1).